Unit 10, Lesson 4: Writing and Solving Absolute Value Equations

Benchmark

MA.912.AR.4.1 Given a mathematical or real-world context, write and solve one-variable absolute value equations.

Learning Targets

- ☐ I can solve multi-step absolute value equations.
- ☐ I can write multi-step absolute value equations.

10.4.1 Warm-Up: Bell Work

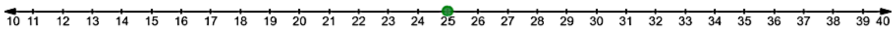
Solve each equation.

1. $|r| + 25 = 39$

2. $|s - 43| = 92$

10.4.2 Collaboration: Writing Absolute Value Equations

1. A number line with a point at 25 is shown.



   a. Place a point at 18 and at 32 and complete the table.

Point	Distance from 25
18	
32	

   b. The absolute value equation, $|x - 25| = 7$, has the solutions $x = 18$ and $x = 32$. Work with your partner to write a description of each quantity in the equation.

	Description
7	
25	
x	

   c. Solve $|x - 25| = 7$.

The solutions to $|x - 25| = 7$ are the minimum and maximum. At times, it is necessary to consider the minimum and maximum of a range.

2. The width of a machine part should be 4.75 centimeters (cm). The manufacturing tolerance is within 0.05 cm.

   a. What is the minimum value the machine part could be?

   b. What is the maximum value the machine part could be?

   c. Sketch a number line that shows the minimum value, maximum value, and 4.75.

   d. Write an absolute value equation that could be used to determine the maximum and minimum value of w, the width of the machine part, algebraically.

   e. Show that the solutions to the equation match the maximum and minimum values determined in Parts A and B.

10.4.3 Guided Instruction: Solving Multi-Step Absolute Value Equations

1. The mass of a regulation baseball should be 145.5 grams (g). The manufacturing tolerance is within 3.5 g.

   a. Write an absolute value equation that could be used to determine the maximum and minimum value of m, the mass of a regulation baseball.

   b. Solve the equation.

   c. Express all possible values of m according to the manufacturing tolerance using set-builder notation.

2. Solve each equation.

   a. $|y-7|-4=3$

   b. $0=|5y-18|-42$

   c. $64=-4|2y+9|$

   d. $|5-8x|+11=11$

   e. Discuss with your partner which, if any, of the equations will have an intersection at the vertex. Summarize your discussion.

10.4.4 Activity: Card Sort

Work with your partner to solve each equation. Use the cards to sort the solutions. All solution cards will be used one time. If all of the solution cards are not used once, then work with your partner to find any mistakes you might have made.

$\lvert x-1\rvert+2=5$	$-\lvert x-1\rvert-1=0$
$\lvert x+1\rvert-2=-2$	$7\lvert 1-x\rvert+3=24$

$\lvert x+1\rvert+3=2$	$-3\lvert x-1\rvert+1=1$
$\lvert 2x+3\rvert-4=1$	$\lvert x-2\rvert+6=6$
$\lvert x+2\rvert+7=4$	$6\lvert x+1\rvert-18=0$

10.4.5 Wrap-Up: Ticket Out the Door

1. Solve each equation.

   a. $0 = 2|x + 4| - 12$

   b. $|-3x + 1| - 8 = 0$

Unit 10, Lesson H1: Solving Absolute Value Inequalities

Benchmark

MA.912.AR.4.2 Given a mathematical or real-world context, write and solve one-variable absolute value inequalities. Represent solutions algebraically or graphically.

Learning Targets

- ☐ I can solve absolute value inequalities.
- ☐ I can represent solutions to absolute value inequalities algebraically or graphically.